COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010	**SECTION:**	CPE21S1
DATE PERFORMED:	July 8, 2026	**DATE SUBMITTED:**	July 9, 2026
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ACTIVITY NO. 1.1 Algorithm, Flowchart, Pseudocode

1. PROCEDURE OUTPUT

Performed basic C++ programming by declaring variables, accepting user input, performing arithmetic operations, and displaying the computed total and average using input and output statements.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

N/A

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

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By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

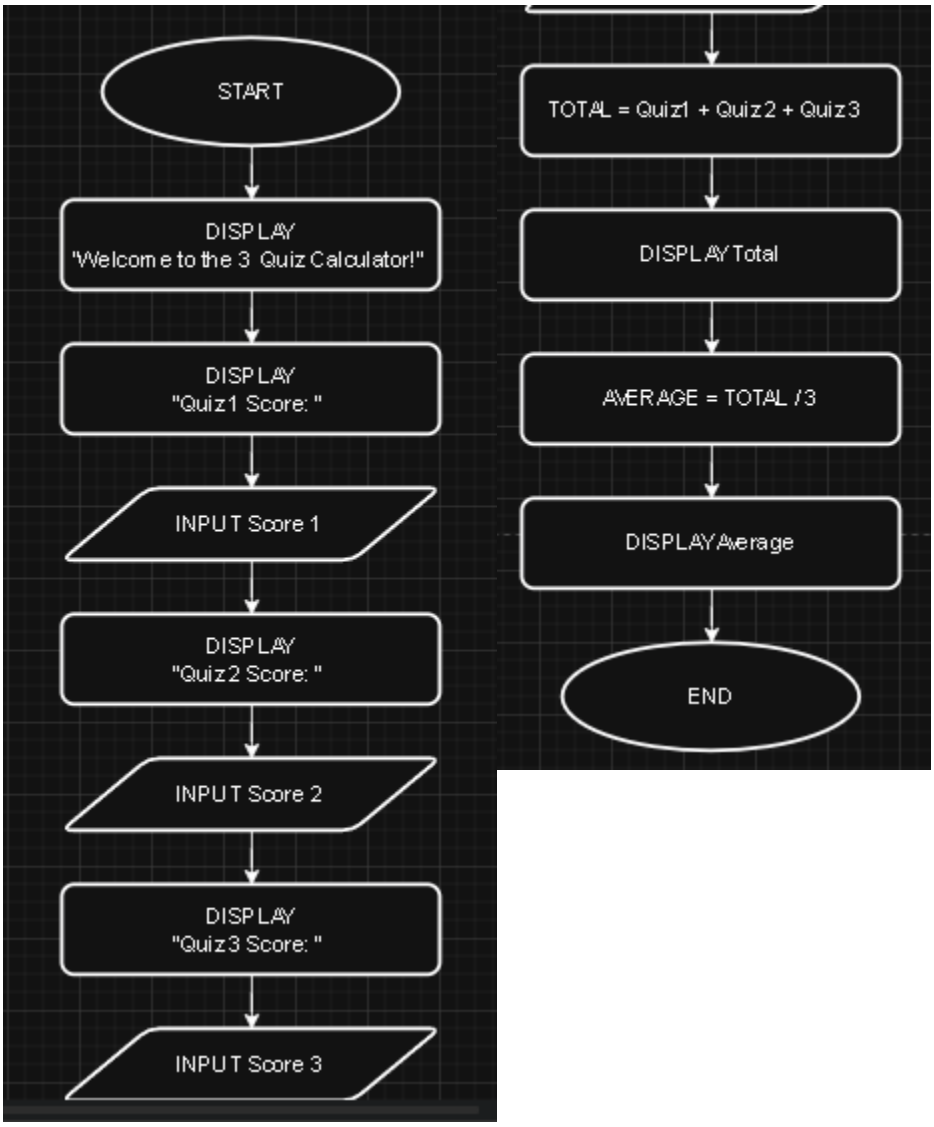
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Pseudocode

START
DISPLAY ""Welcome to the 3 Quiz Calculator!"
DISPLAY "Enter Quiz 1 Score:"
INPUT Score 1
DISPLAY "Enter Quiz 2 Score:"
INPUT Score 2
DISPLAY "Enter Quiz 3 Score:"
INPUT Score 3
TOTAL = Quiz1 + Quiz 2 + Quiz 3
DISPLAY Total Grade
AVERAGE = Total / 3
DISPLAY Average
END

Flowchart



Code

SeatworkNumber1.cpp ● SeatworkNumber2.cpp

SeatworkNumber1.cpp > ...

```cpp
#include <iostream>
using namespace std;

int main()
{
    float quiz1, quiz2, quiz3;
    float total, average;

    cout << "Enter Quiz 1: ";
    cin >> quiz1;

    cout << "Enter Quiz 2: ";
    cin >> quiz2;

    cout << "Enter Quiz 3: ";
    cin >> quiz3;

    total = quiz1 + quiz2 + quiz3;
    average = total / 3;

    cout << "\nTotal = " << total << endl;
    cout << "Average = " << average << endl;

    return 0;
}
```

Code Output

```
PS C:\Users\mendo\OneDrive\Documents\C++\output> & .\'SeatworkNumber1.exe'
Welcome to the 3 Quiz Calculator!
Enter Quiz 1: 10
Enter Quiz 2: 20
Enter Quiz 3: 30

Total = 60
Average = 20
```

Code Analysis: The program accepts three quiz scores from the user, calculates their total and average, and then displays the results.

Pseudocode

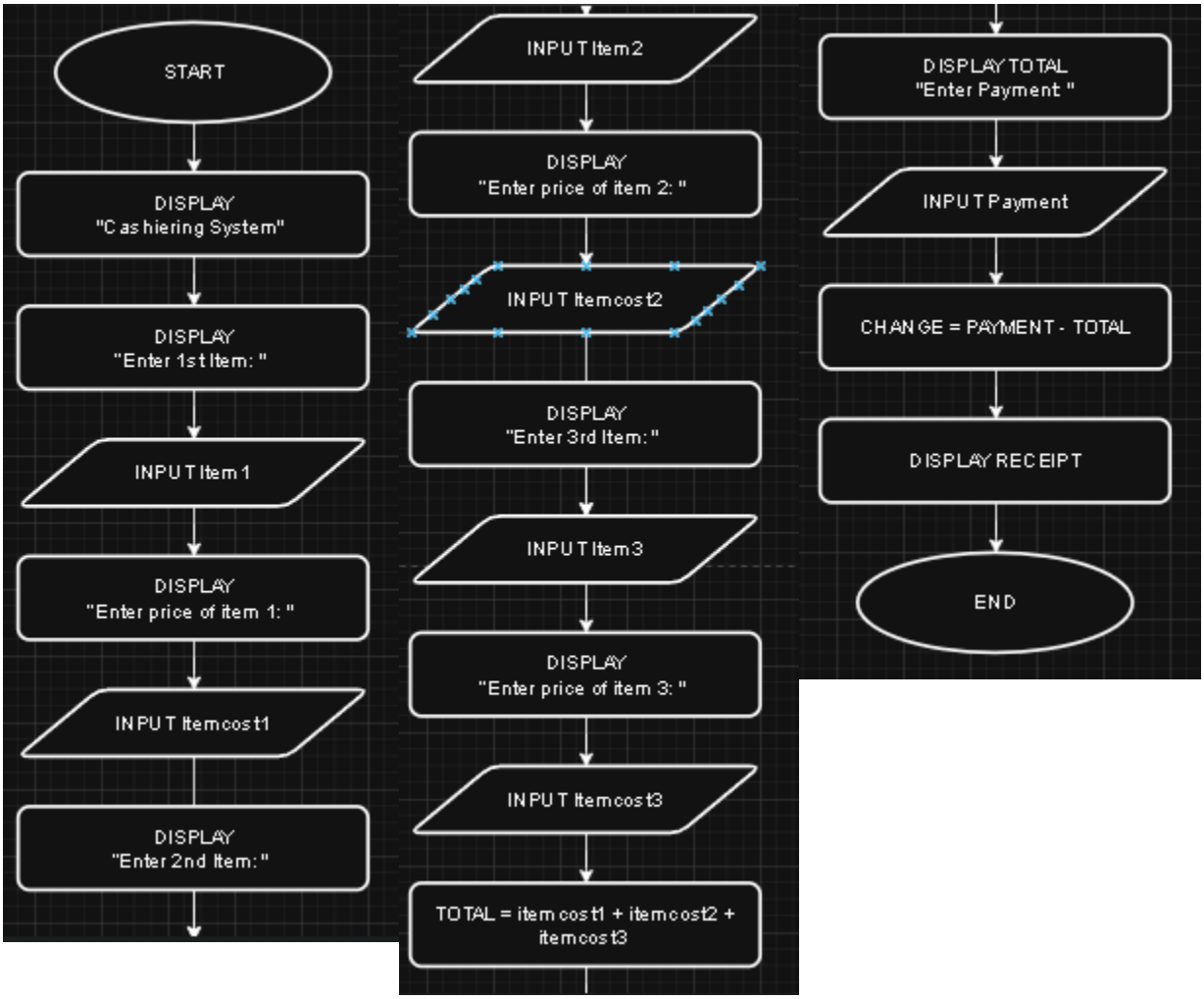
START
- DISPLAY "Cashiering System"
- DISPLAY "Enter 1st Item: "
- INPUT Item1
- DISPLAY "Enter item1 Cost"
- INPUT Itemcost1
- DISPLAY "Enter 2nd Item: "
- INPUT Item2
- DISPLAY "Enter item2 Cost"
- INPUT Itemcost2
- DISPLAY "Enter 3rd Item: "
- INPUT Item3
- DISPLAY "Enter item3 Cost"
- INPUT Itemcost3
- TOTAL = itemcost1 + itemcost2 + itemcost3
- DISPLAY TOTAL
- INPUT PAYMENT
- CHANGE = PAYMENT - TOTAL
- DISPLAY RECEIPT

END

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Flowchart



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Code

SeatworkNumber1.cpp | SeatworkNumber2.cpp

SeatworkNumber2.cpp > main()

```cpp
#include <iostream>
#include <string>
using namespace std;

int main() {
    string item1, item2, item3;
    double itemcost1, itemcost2, itemcost3;
    double total, payment, change;

    cout << "===== Cashiering System =====\n";

    cout << "Enter 1st Item: ";
    cin >> item1;

    cout << "Enter price of Item 1: ";
    cin >> itemcost1;

    cout << "Enter 2nd Item: ";
    cin >> item2;

    cout << "Enter price of Item 2: ";
    cin >> itemcost2;
```

```
25      cin >> item3;
26
27      cout << "Enter price of Item 3: ";
28      cin >> itemcost3;
29
30      total = itemcost1 + itemcost2 + itemcost3;
31
32      cout << "\nTotal: " << total << endl;
33
34      cout << "Enter payment: ";
35      cin >> payment;
36
37      change = payment - total;
38
39      cout << "\n----- Receipt -----\n";
40      cout << "Item 1: " << item1 << " P" << itemcost1 << endl;
41      cout << "Item 2: " << item2 << " P" << itemcost2 << endl;
42      cout << "Item 3: " << item3 << " P" << itemcost3 << endl;
43      cout << "Total: " << total << endl;
44      cout << "Payment: " << payment << endl;
45      cout << "Change: " << change << endl;
```

```
46
47      return 0;
48  }
49
```

Code Output

```
===== Cashiering System =====
Enter 1st Item: Apple
Enter price of Item 1: 100
Enter 2nd Item: Oil
Enter price of Item 2: 300
Enter 3rd Item: Potato
Enter price of Item 3: 400

Total: 800
Enter payment: 1000
```

```
----- Receipt -----
Item 1: Apple P100
Item 2: Oil P300
Item 3: Potato P400
Total: 800
Payment: 1000
Change: 200
```



Code Analysis: This program prompts the user twice per item for the cost and its name so that the program will be able to print a receipt. Additionally, the program also calculates the total cost and asks the user for payment and calculates the change.

Conclusion

This activity brought me back to my 1st year days wherein it was my first ever programming language experience. I am really glad that I still remember the basics which allows me to complete this activity in the laboratory which serves as a good refresher on basics. Even though, its not really that deep coding where it will use midterm techniques on programming logic on design such as using function and etc. The program is still executable in basic form.